Mini Project Worksheet: Rectangular Microstrip Patch Antenna (RMA) Design

Course: EE5606 Artificial Intelligence for Antennas in Wireless Communication

Report Submission Deadline: Week 12 (at the commencement of the lecture)

Project Background

Minimize the effective volume of a rectangular microstrip patch antenna (RMA) ($V = W \cdot L \cdot h$) operating at 2.4 GHz. Traditional full-wave simulations (e.g., HFSS/CST) are time-consuming. For simplicity, no full-wave simulations are required in this project. You are required to do the following two parts:

Part I: Use an analytical formula (Transmission-Line Model) to calculate the RMA resonance frequency and optimize the RMA dimensions (patch width W, patch length L, and substrate thickness h) using the genetic algorithm (GA).

Part II: Use a surrogate model to model the frequency formula. Practically, a surrogate model is used to replace time-consuming electromagnetic simulations with a software package (e.g., HFSS/CST).

Part I: Optimization of RMA using Genetic Algorithm

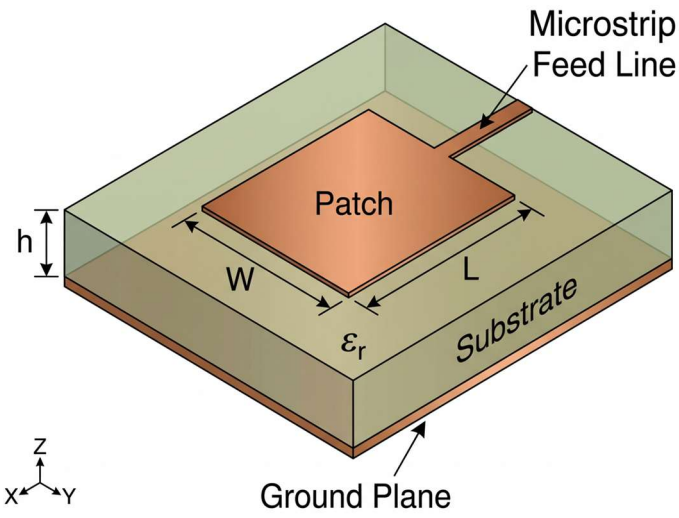


Fig. 1

Design Goal:

Minimize the effective volume of the RMA ($V = W \cdot L \cdot h$) operating at $f_0 = 2.4$ GHz.

Objective Function:

Define an objective function for the optimization (combining the volume V and a penalty term for the frequency deviation $|f$ - 2.4 GHz |.

Constant Parameter:

Relative dielectric constant of the substrate $\varepsilon_r = 4.4$ (FR-4 substrate, speed of light $c = 3 \times 10^8$ m/s).

Design Variables & Search Space: Size of the RMA:

- Patch width W in [25, 50] mm
- Patch length L in [25, 45] mm
- Substrate thickness h in [1.0, 5.0] mm

(Note: To ensure practical radiation efficiency and avoid degenerate thin/narrow geometries, enforce the geometrical constraint $W / L \geq 1.2$ during optimization).

Formulas:

Use the following transmission-line model formulas to calculate the dominant TM_{010}-mode resonance frequency f of the RMA:

$$f = \frac{c}{2L_{\text{eff}}\sqrt{\varepsilon_{\text{eff}}}}$$

where the effective dielectric constant ε_{eff} is given by:

$$\varepsilon_{\text{eff}} = \frac{\varepsilon_{\text{r}} + 1}{2} + \frac{\varepsilon_{\text{r}} - 1}{2}\left(1 + 12\frac{h}{W}\right)^{-1/2}$$

and the effective length L_{eff}, accounting for the fringing fields ΔL at both ends of the patch, is given by:

$$L_{\text{eff}} = L + 2\Delta L$$

$$\Delta L = 0.412h\frac{\left(\varepsilon_{eff} + 0.3\right)\left(\frac{W}{h} + 0.264\right)}{\left(\varepsilon_{eff} - 0.258\right)\left(\frac{W}{h} + 0.8\right)}$$

Part II: Use Surrogate Model to Accelerate Optimization

In this part, you will train a surrogate model to approximate the frequency formula that govern the RMA's resonance frequency:

Data Generation: Generate a dataset using sampling methods in the design space (W, L, h).

Model Training: Train a reliable surrogate model to predict the operating frequency f.

Optimization: Use the trained model with the particle swarm optimization (PSO) to solve the RMA design problem.

Verification: Use the frequency formula to verify your result.

Note: Students are required to compare the results of **2 surrogate models (NN, and Gaussian Process)**.

Submission Requirements

Upload your ZIP file to "Assignments" in Canvas. The ZIP file should contain:

1. **Code:** Provide comprehensive comments for each part (Data generation, model training, Optimization algorithm).
2. **Brief Report:** Including methodology, results, discussion, and conclusion (no more than 4 pages, single line spacing, Times New Roman font 12, 1-inch margin from top, bottom, left, and right).
3. **Presentation (8 min + 2 min Q&A):** Including a demo video.
4. **Demo video**: including the training processes and optimization of part II.
5. **Note:** In your presentation slides, use diagrams, plots, figures, etc. as far as possible. Avoid long paragraphs and keep the text as concise as possible.
6. **Suggestion:**
    - Use flowcharts to explain your algorithm workflows.
    - Include plots to visualize algorithm performance (e.g., convergence curves, prediction ability).
    - Use a table for the comparison between the 2 surrogate models.

Grading Criteria

Category	Weighting	
Code Quality	25%	Readability, modularity, and documentation
Algorithm Implementation	25%	Correctness of optimization and ML methods
Report	25%	Organization, conciseness, and analysis
Presentation	25%	Clarity and time management